

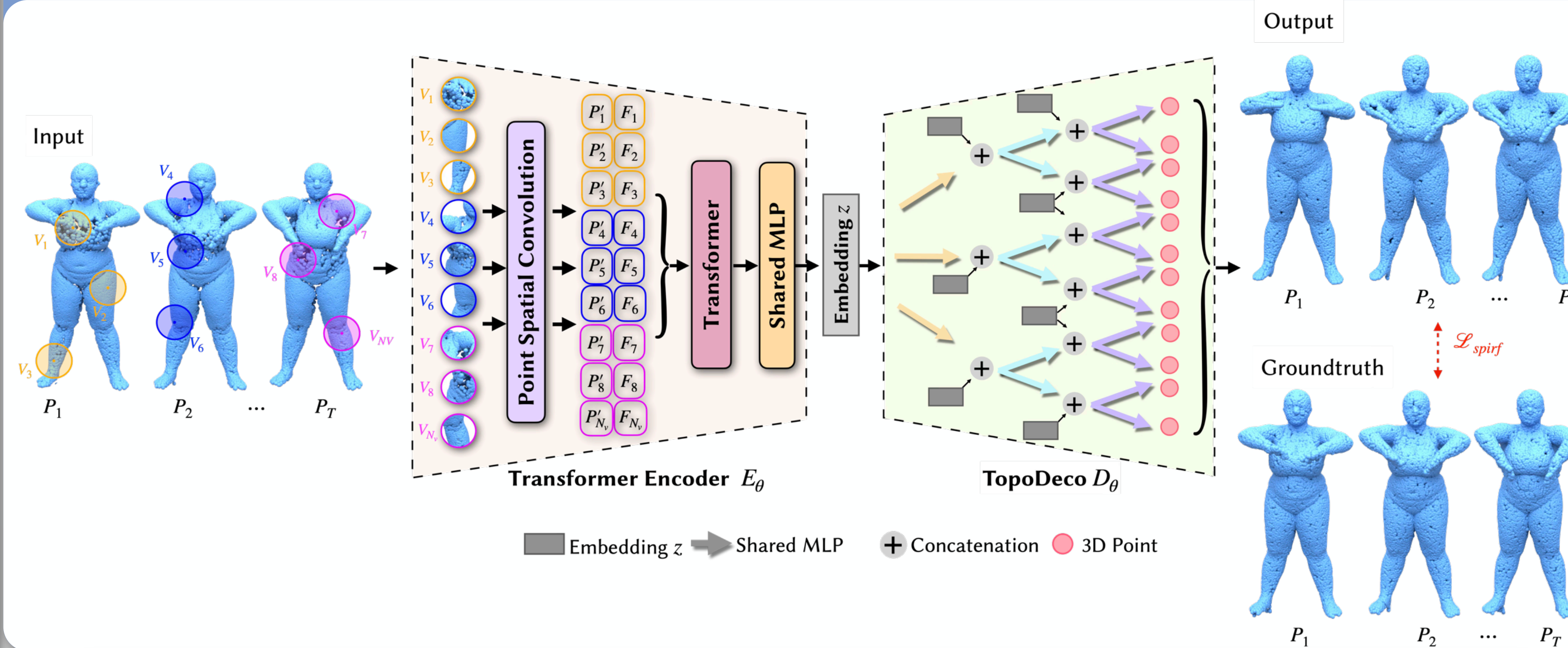
## Abstract

Reconstructing high-fidelity 4D human point clouds from incomplete scans is challenged by complex topologies, rapid motion, and real-world sensor occlusions. Because traditional frame-independent methods often cause temporal discontinuities and structural distortions, we propose SPiRF4Humans, a spatio-temporal refinement framework for 4D human capture. Our approach features a spatio-temporal transformer encoder and a topology-aware hierarchical decoder to reconstruct missing regions by learning geometric relationships across the sequence. To preserve structural integrity, we introduce a DGCNN-inspired perceptual loss that penalizes high-dimensional structural discrepancies. Additionally, we simulate realistic occlusions using a physically-grounded synthetic data generation strategy based on multi-view ray-casting and asymmetric sampling. Experimental results demonstrate that SPiRF4Humans outperforms state-of-the-art methods, yielding the geometrically coherent and temporally stable reconstructions necessary for motion analysis, animation, and XR applications.

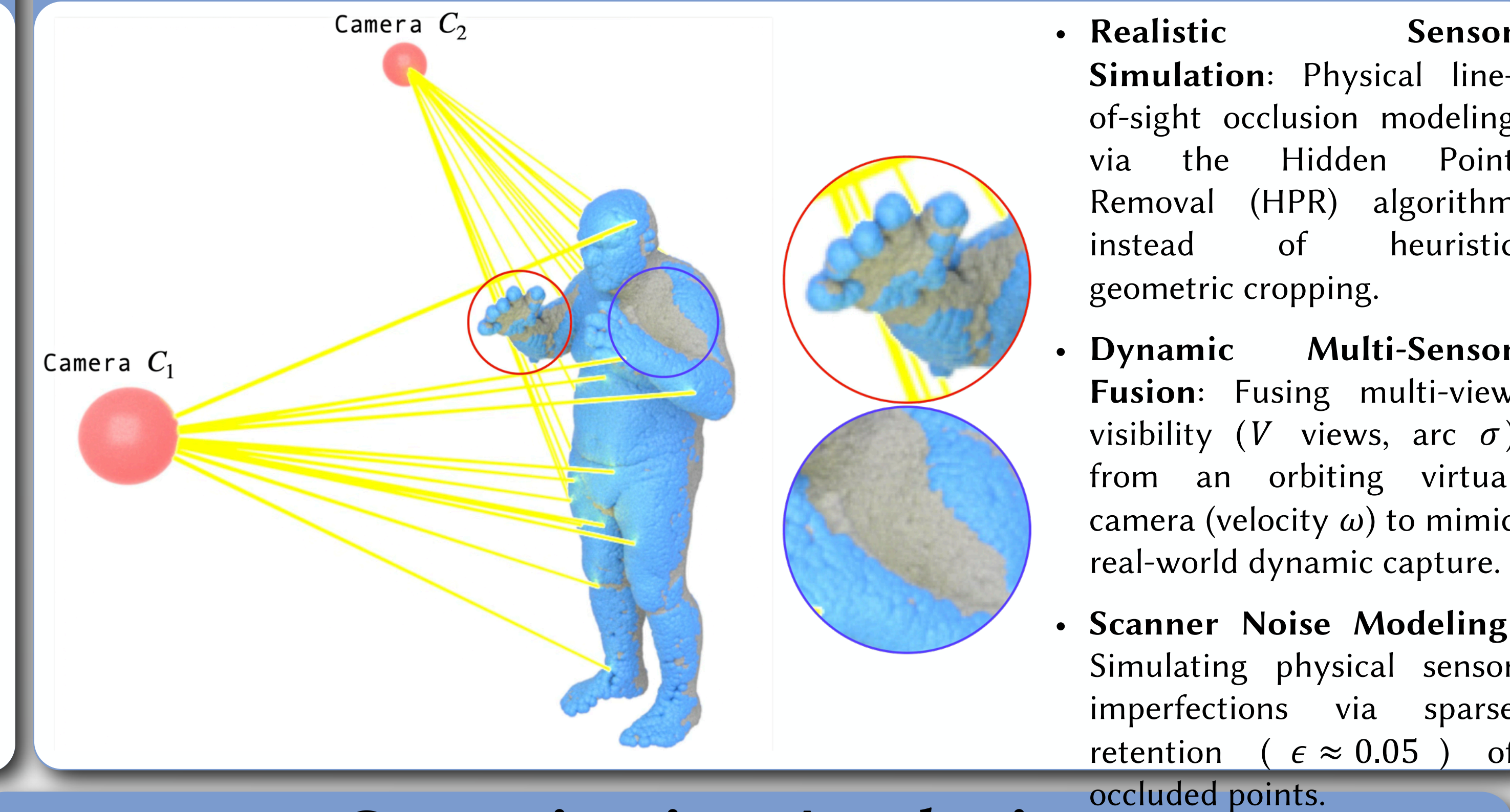
## Contributions

- **Spatio-Temporal Transformer Encoder:** Captures dynamic motion features using point spatial convolutions and 4D space-time coordinate embeddings.
  - **Topology-Aware Tree Decoder (Topo-Deco):** Reconstructs complex, time-varying human manifolds progressively using a hierarchical tree structure with shared MLPs.
  - **Dynamic Graph Perceptual Loss:** Supervises shape refinement in high-dimensional feature space using pre-trained DGCNN features to preserve spatial and temporal information
- **Dynamic Occlusion Dataset:** Simulates realistic line-of-sight sensor occlusions via multi-view HPR ray-casting on MPI D-FAUST 4D scans.

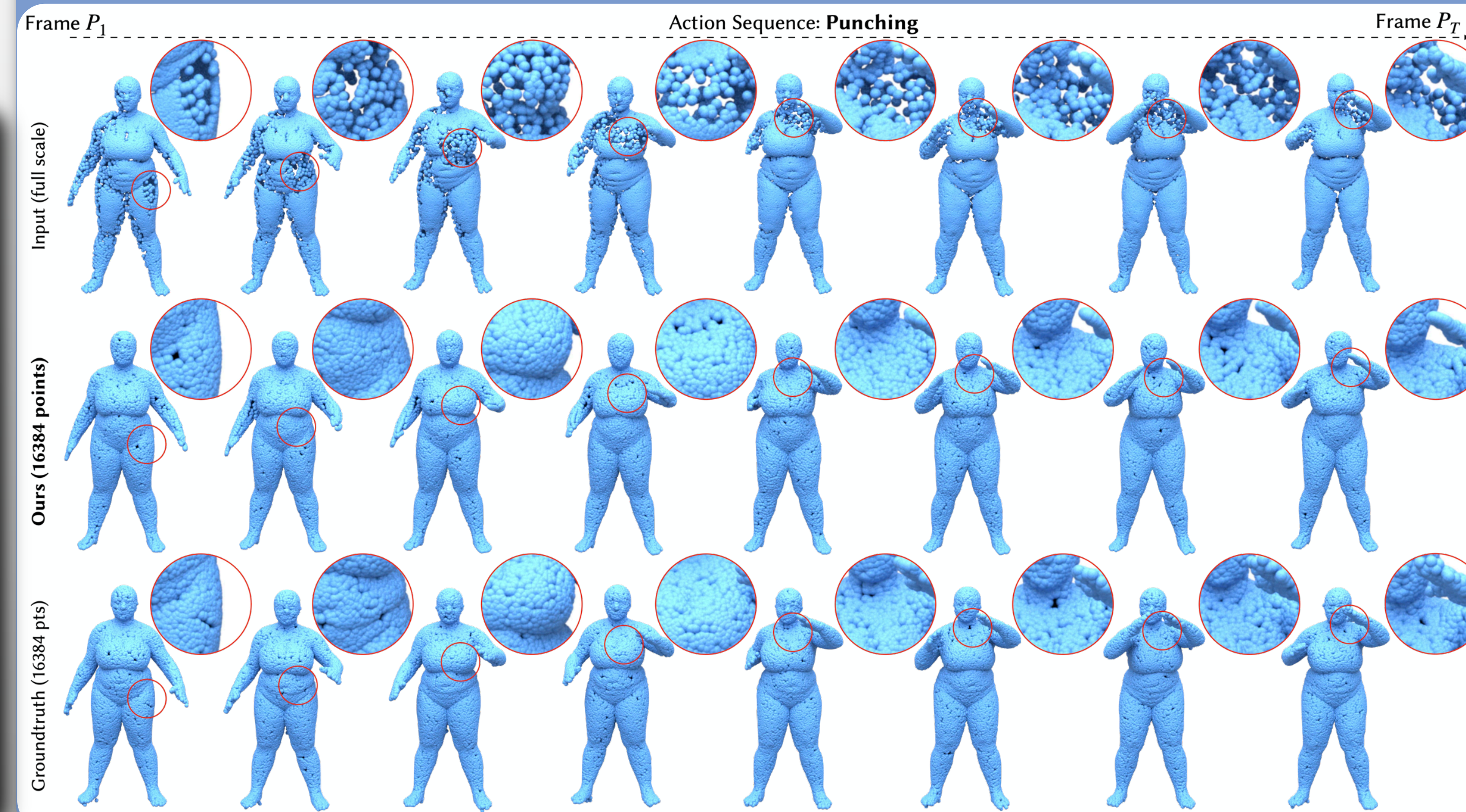
## Proposed SPiRF4Humans Method



## Synthetic Data Generation



## Results: 4D Point Cloud Refinement



## Quantitative Analysis

Table 1. Quantitative comparison of PCN, SPiKE, and SPiRF4Humans using Chamfer Distance (CD), Earth Mover's Distance (EMD), and our proposed  $L_{spirf}$  loss on 4D action sequences. While PCN fails to capture temporal coherence, SPiKE demonstrates substantial improvements. Our method achieves significantly higher fidelity and surface stability. **Note:** † indicates that the baseline was adapted from its original task to generative 4D completion via a coordinate-based decoder and temporal regularization.

| Action Sequences | PCN† [55]     |               | SPiKE† [1]    |               | SPiRF4Humans (Ours) |               |                        |
|------------------|---------------|---------------|---------------|---------------|---------------------|---------------|------------------------|
|                  | CD ↓          | EMD ↓         | CD ↓          | EMD ↓         | CD ↓                | EMD ↓         | $L_{spirf} \downarrow$ |
| Chicken Wings    | 8.6802        | 4.0281        | 7.5210        | 3.8102        | 1.0929              | 0.0242        | 0.000144               |
| Hips             | 7.8520        | 4.1101        | 6.9123        | 3.9021        | 0.4421              | 0.0210        | 0.000140               |
| Jiggle On Toes   | 5.9127        | 3.2982        | 5.1021        | 3.0124        | 1.0990              | 0.0305        | 0.000208               |
| Jumping Jacks    | 4.3210        | 2.9202        | 3.8921        | 2.5012        | 0.7099              | 0.0288        | 0.000297               |
| Knees            | 2.8320        | 1.0211        | 2.1023        | 0.9102        | 0.3010              | 0.0156        | 0.000147               |
| Light Hopping    | 8.4022        | 4.3025        | 7.6102        | 4.1023        | 1.4273              | 0.0311        | 0.000137               |
| One Leg Jump     | 6.4867        | 2.3900        | 5.9012        | 2.1023        | 1.1607              | 0.0225        | 0.000156               |
| One Leg Loose    | 7.7609        | 4.9022        | 6.8123        | 4.5012        | 2.1789              | 0.0345        | 0.000165               |
| Punching         | 9.3630        | 5.8492        | 8.4102        | 5.2012        | 1.2134              | 0.0412        | 0.000174               |
| Running On Spot  | 1.2865        | 0.2920        | 1.1023        | 0.2501        | 2.2357              | 0.0102        | 0.000228               |
| Shake Arms       | 9.1625        | 5.3910        | 8.2102        | 5.0123        | 1.2905              | 0.0389        | 0.000153               |
| Shake Hips       | 7.9229        | 4.3902        | 7.1023        | 4.0123        | 0.8332              | 0.0322        | 0.000131               |
| Shake Shoulders  | 8.5087        | 5.2933        | 7.7123        | 4.9021        | 0.9317              | 0.0367        | 0.000146               |
| <b>Average</b>   | <b>6.8380</b> | <b>3.7067</b> | <b>6.0321</b> | <b>3.3331</b> | <b>1.1474</b>       | <b>0.0282</b> | <b>0.000171</b>        |

## References

- Bogu, Federica, et al. "Dynamic FAUST: Registering human bodies in motion." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.
- Ballester, Irene, Ondřej Peterka, and Martin Kampel. "Spike: 3d human pose from point cloud sequences." International Conference on Pattern Recognition. Springer, Cham, 2025.

